Memory Notes

Simplified DDR3 Simplified edition - JEDEC JESD79-3F

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Table of Contents

1 Simplified Ddr3 Index....10
2 Introduction....11
2.1 What DDR3 SDRAM is....11
2.2 Key features of DDR3....12
2.3 Densities and speed grades....13
2.4 The simplified drill model....14
3 Organization....15
3.1 Banks, rows, columns....15
3.2 Addressing....16
3.3 Prefetch and IO width....17
3.4 Ranks and stacked packages....18
3.5 The drill model restated....19
4 Acronyms and Terms....20
5 Power-Up and Initialization....23
5.1 Cold start sequence....24
5.2 Reset with stable power....26
5.3 Notes that bite....27
6 Mode Registers....28
6.1 MR0 (BA2=BA1=BA0=0)....29
6.2 MR1 (BA2=0, BA1=0, BA0=1)....31
6.3 MR2 (BA2=0, BA1=1, BA0=0)....32
6.4 MR3 (BA2=0, BA1=1, BA0=1)....34

6.5 Programming rules....35
7 Command Truth Tables....36
7.1 Command truth table....37
7.2 CKE truth table....41
7.3 Data mask....42
7.4 Not in the table....43
8 Commands: Rules and Constraints....44
8.1 ACT - Bank Activate....45
8.2 RD / RDA - Read....46
8.3 WR / WRA - Write....47
8.4 PRE / PREA - Precharge....48
8.5 REF - Refresh....49
8.6 Self-refresh entry / exit....50
8.7 Power-down....51
8.8 NOP and DES....52
8.9 RESET....53
8.10 MRS / mode-register updates....54
8.11 Frequency change in precharge power-down....55
8.12 Write leveling entry / exit....56
8.13 MPR read flow....57
9 Walkthrough 1: AXI4 Read, Row Hit....58
9.1 Controller configuration used in all Chapter 4 walkthroughs....58
9.2 Scenario....59
9.3 Command trace....60

9.4 What happens on the AXI side....61
9.5 Scheduling takeaways....62
10 Walkthrough 2: AXI4 Read, Row Miss....63
10.1 Scenario....64
10.2 Command trace....65
10.3 Cost accounting....66
10.4 AXI side....67
10.5 Variation: read with auto precharge (RDA)....68
11 Walkthrough 3: AXI4 Write to an Idle Bank....69
11.1 Scenario....70
11.2 Command trace....71
11.3 AXI side....72
11.4 Write recovery is the trap....73
11.5 Comparison against the read miss....74
12 Walkthrough 4: Mixed Traffic and Bus Turnarounds....75
12.1 Scenario....76
12.2 Command trace....77
12.3 The two turnaround rules....78
12.4 Scheduling takeaways....79
12.5 AXI side....80
13 Data Path: Prefetch, Bursts, Strobes....81
13.1 8n prefetch....81
13.2 Burst length and burst order....82
13.3 DQS: the source-synchronous strobe....84

13.4 Write leveling....85
13.5 Write data mask (DM)....86
13.6 Latency arithmetic, collected....87
14 ODT - On-Die Termination....88
14.1 What it is....88
14.2 Programming....89
14.3 ODT timing....90
14.4 System rules....91
14.5 ZQ calibration....92
14.6 What DDR3 does NOT have....93
15 Core Timing Parameters....94
15.1 Row-cycle parameters....95
15.2 Activate-rate parameters (cross-bank)....96
15.3 Data-to-precharge parameters....97
15.4 Why these four numbers dominate....98
16 Turnaround and Burst Timing....99
16.1 CAS-to-CAS and latencies....100
16.2 Direction changes....101
16.3 Strobe-level timing (for completeness)....102
16.4 Burst chop and interrupt rules....103
17 Refresh and Self-Refresh Timing....104
17.1 Refresh parameters....104
17.2 Self-refresh....105
17.3 Power-down timings (related CKE rules)....107

18 Package and Electrical Notes....108
18.1 Packages....109
18.2 Supplies and signaling....110
18.3 Speed bins....111
18.4 Characterization....112
19 Test and Calibration Features....113
19.1 Write leveling....114
19.2 Multi-Purpose Register (MPR)....116
19.3 ZQ calibration....117
19.4 Self-refresh temperature (SRT / ASR)....118
19.5 Test mode....119
19.6 What is not here....120

List of Figures

No figures in this document.

List of Tables

No tables in this document.

List of Waveforms

No waveforms in this document.

1 Simplified Ddr3 Index

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2 Introduction

2.1 What DDR3 SDRAM is

DDR3 SDRAM is the third generation of double-data-rate synchronous DRAM, standardized by JEDEC as JESD79-3F. For every column access it pulls eight data words from the memory array (8n prefetch) and then transfers two words per clock cycle at each DQ pin, so the internal array runs at roughly one eighth of the data rate seen at the pins. The nominal supply is 1.5 V (SSTL_15), a step down from DDR2's 1.8 V.

Family positioning, in one line each:

Tech	Prefetch	Distinctives
SDR SDRAM	1n	Single data rate, one word per clock
DDR SDRAM	2n	Double data rate, 2.5 V SSTL_2
DDR2 SDRAM	4n	ODT, posted CAS / additive latency, 1.8 V SSTL_1.8
DDR3 SDRAM	8n	1.5 V, fly-by CA topology, ZQ calibration, write leveling
DDR4 SDRAM	8n	Bank groups, lower voltage, higher data rates

2.2 Key features of DDR3

- 8n prefetch: every column access moves eight words internally.
- Burst length 8 (BL8) or burst chop 4 (BC4). BC4 can be chosen on-the-fly through A12/BC# during a read or write, or fixed in the mode register.
- Posted CAS with additive latency (AL) is retained. A read or write can be issued immediately after activate and is held inside the DRAM for AL clocks. Read latency RL = AL + CL; write latency WL = AL + CWL (not RL - 1 as in DDR2).
- The DLL is required and enabled during initialization; it aligns DQS to CK for reads and supports write leveling.
- A dedicated RESET# pin is new compared with DDR2. It performs an asynchronous hardware reset.
- ZQ calibration uses the external ZQ reference pin to set output driver impedance and on-die termination value.
- Dynamic ODT (Rtt_WR) lets the DRAM switch to a separate termination strength during writes.
- Write leveling lets each DRAM rank align its DQS output to the incoming CK/CK# in a fly-by topology.
- The multi-purpose register (MPR) is a read-only register used for read leveling and DQ eye training.
- Off-chip driver (OCD) calibration from DDR2 is removed; impedance calibration is now handled through ZQ calibration.
- No DBI, no parity, no on-die ECC. Data integrity is left to the controller and the rest of the system.
- Flat eight-bank organization on every standard density; bank groups are a later-generation concept.
- Refresh is all-bank only (no per-bank refresh command).
- Self-refresh includes a self-refresh temperature (SRT) option for extended temperature range.

2.3 Densities and speed grades

Standard monolithic densities run from 512 Mb to 8 Gb in x4, x8 and x16 organizations. Every standard density uses eight banks selected by BA0-BA2.

Speed grades are named by data rate: DDR3-800, DDR3-1066, DDR3-1333, DDR3-1600, DDR3-1866 and DDR3-2133. Each grade carries a letter suffix that identifies the CL-tRCD-tRP bin (for example, DDR3-800E is a 6-6-6 bin).

2.4 The simplified drill model

Real parts have thousands of rows and over a thousand columns per row. All examples in this book use a deliberately tiny geometry so bank state is obvious at a glance:

- Banks: B0-B7 (eight, as on all standard DDR3 devices)
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

This model is used by the ddr_drills training app. Every command trace, walkthrough and timing example in later chapters uses it. Real address maps are far larger; the scheduling rules are identical.

Source: JESD79-3F sections 1, 2.11, 3.2

3 Organization

3.1 Banks, rows, columns

A DDR3 device is a flat set of eight independent banks. Each bank is a 2D array addressed by row then column. An Activate command copies one row into that bank's sense amplifiers (the open row or page); reads and writes then hit the sense amplifiers at column granularity. A bank must be precharged before a different row in the same bank can be opened.

All standard densities use eight banks selected by BA0-BA2:

Density	Org	Row address	Column address	Page size
512 Mb	x8	A0-A12	A0-A9	1 KB
512 Mb	x16	A0-A11	A0-A9	2 KB
1 Gb	x8	A0-A13	A0-A9	1 KB
1 Gb	x16	A0-A12	A0-A9	2 KB
2 Gb	x8	A0-A14	A0-A9	1 KB
4 Gb	x16	A0-A14	A0-A9	2 KB
8 Gb	x8	A0-A15	A0-A9, A11	2 KB

There are no bank groups in DDR3. All banks share one timing regime: tRRD applies between any two Activates, tCCD between any two column commands, and tFAW limits any rolling window of four Activates.

3.2 Addressing

The row address is supplied with the Activate command on A0-A15 (the exact width depends on density). The column address is supplied with the Read or Write command. A10 is not a column bit; it is the auto-precharge flag (AP). A12/BC# chooses between BL8 and BC4 on-the-fly during a column command when OTF is enabled; otherwise the burst mode is fixed in the mode register.

Page size formula: page size = 2^COLBITS x ORG / 8 bytes, per bank. The extra column address bits used on x4 and some x8 densities (A11, and A13 on 8 Gb x4) keep the per-bank page at either 1 KB or 2 KB.

3.3 Prefetch and IO width

The 8n prefetch means one column access pulls eight words per DQ pin group from the array in parallel. Consequences:

- The smallest useful burst is four data words per pin (BC4); BL8 uses the full eight-word fetch.
- The array core runs at one eighth of the pin data rate.
- A BC4 burst transfers the first four words of the prefetch and discards the second four.

IO organizations are x4, x8 and x16. x4 uses one DQS pair for four DQ pins; x8 uses one DQS pair for eight DQ pins; x16 has two DQS pairs, one for each byte lane. x8 parts also support an optional termination data strobe pair, TDQS/TDQS#, when enabled in MR1.

3.4 Ranks and stacked packages

The spec covers single-die, dual-die and quad-die packages. A stacked package presents multiple ranks behind one channel: each rank gets its own chip select (and its own ODT and CKE in multi-rank ballouts), shares the address/command and data busses, and is otherwise a complete independent DDR3 device. Rank-to-rank traffic adds bus-turnaround constraints beyond single-rank rules.

3.5 The drill model restated

This book's examples use eight banks (B0-B7), eight rows per bank (R0-R7) and eight columns per row (C0-C7). One bank's state at a glance:

Bank	State	Open row
B0	idle	-
B3	active	R5
B5	active	R1

Source: JESD79-3F sections 2.10, 2.11, 3.2

4 Acronyms and Terms

Terms used with precise meanings throughout this book.

Term	Meaning
ACT	Bank Activate command: opens a row in a bank
AL	Additive latency: clocks a posted CAS is held inside the DRAM (MR1)
AP	Auto precharge flag, carried on address bit A10 with RD/WR
BA	Bank address pins (BA0-BA2)
BC4	Burst chop 4: a four-word burst from the first half of the 8n prefetch
BL	Burst length: 8 words (BL8) or 4 words via burst chop (BC4)
CAS	Column Address Strobe (command pin); shorthand for a column command
CKE	Clock enable; gates power-down and self-refresh entry/exit
CL	CAS latency: read-command-to-data delay after the posted hold (MR0)
CWL	CAS write latency: write-command-to-first-DQS delay (MR2)
DES	Deselect command (CS# high)
DLL	Delay-locked loop aligning read DQS to CK
DM	Data mask pin (write data masking, per byte)
DQS	Data strobe; source-synchronous clock for DQ in both directions
MPR	Multi-purpose register: read-only register used for training
MR / MRn	Mode Register n (0, 1, 2, 3)

Term	Meaning
MRS	Mode Register Set command
NOP	No Operation command
ODT	On-die termination (Rtt on DQ/DQS/DM, per MR1)
OTF	On-the-fly: A12/BC# selects BL8 or BC4 during a column command
PRE	Precharge command (A10=0: one bank; A10=1: all banks)
RD / WR	Read / Write commands (A10=0)
RDA / WRA	Read / Write with auto precharge (A10=1)
REF	Refresh command (auto refresh, one internal row per bank)
RESET#	Active-low asynchronous reset pin
RL	Read latency = AL + CL
Rtt_NOM	Nominal on-die termination strength
Rtt_WR	Dynamic ODT strength used during writes
SRT	Self-refresh temperature option (MR2)
SRE / SRX	Self-refresh entry / exit
SSTL_15	1.5 V stub-series-terminated logic IO standard
tCK	Clock period
TDQS / TDQS#	Optional termination data strobe pair (x8 only, MR1)
WL	Write latency = AL + CWL
ZQ	Reference pin for ZQ calibration
ZQCL / ZQCS	ZQ calibration long / short commands
Bank group	A DDR4 concept; does not exist in DDR3
Page / open row	The row currently held in a bank’s sense amplifiers

Command trace notation used in Chapters 3 and 4:

```
ACT B3 R5      # open row R5 in bank B3
RD  B3 C2      # read burst starting at column C2
WR  B1 C0      # write burst starting at column C0
RDA B2 C4      # read with auto precharge
PRE B3         # precharge bank B3
PREA           # precharge all banks
REF            # auto refresh
ZQCS / ZQCL    # ZQ calibration short / long
--- (tWTR bubble) ---   # forced idle bus time, reason named
```

Source: JESD79-3F sections 2.10, 2.11, 3.1, 3.2

5 Power-Up and Initialization

DDR3 must be brought up through a fixed initialization flow; deviating from it produces undefined behavior. The flow below covers both cold power-up and a reset while power remains stable.

5.1 Cold start sequence

1. Apply power with RESET# held below 0.2 x VDD. Other input levels are allowed to be undefined during this phase. RESET# must stay low for at least 200 us once the supplies are stable, and CKE must be driven low at least 10 ns before RESET# rises.
2. Keep the voltage ramp within the required windows: from 300 mV up to VDDmin in no more than 200 ms, with VDD at or above VDDQ and the difference between them under 0.3 V. Either use a single regulator for VDD/VDDQ (with all non-power pins bounded between VSS/VSSQ and VDD/VDDQ, VTT clamped to 0.95 V max after ramp, and VREF following VDDQ/2), or ramp VDD no later than VDDQ and VDDQ no later than VTT and VREF while avoiding slope reversals.
3. Release RESET#, then wait 500 us before raising CKE. During this window the DRAM performs internal state initialization without needing a clock.
4. Start CK/CK# and let it stabilize for at least 10 ns or 5 tCK, whichever is longer, before CKE goes high. A NOP or Deselect command must be registered before the CKE rising edge, and CKE must stay high without interruption for the rest of initialization, through tDLLK and tZQinit.
5. Keep ODT high-impedance from power-up until CKE is high. If RTT_NOM is enabled in MR1, hold the ODT pin low from tIS before CKE rises until the end of initialization; otherwise it may be tied low or high but must not toggle.
6. After CKE is registered high, wait tXPR = max(tXS, 5 x tCK) before the first MRS command.
7. Issue MRS to load MR2 (BA2=0, BA1=1, BA0=0).
8. Issue MRS to load MR3 (BA2=0, BA1=1, BA0=1).
9. Issue MRS to load MR1 with the DLL enabled (A0=0; BA2=0, BA1=0, BA0=1).
10. Issue MRS to load MR0 and reset the DLL (A8=1; BA2=0, BA1=0, BA0=0).
11. Issue ZQCL to begin ZQ calibration.
12. Wait until both tDLLK and tZQinit have expired; then normal operation may begin.

The required gaps look like this:

```
... CKE high ... MRS(MR2) ... MRS(MR3) ... MRS(MR1) ... MRS(MR0) ...
ZQCL ... ready
        |tXPR| tMRD | tMRD |    tMRD    |  tMOD  |     tZQinit     |
```

NOP/Deselect must fill every gap from the first MRS through ZQCL.

5.2 Reset with stable power

If power is already stable and only a reset is required, the sequence is shorter:

1. Drive RESET# below 0.2 x VDD for at least 100 ns. CKE must be low at least 10 ns before RESET# rises.
2. Continue from cold-start step 2 (500 us wait, clock stabilization, CKE high, and so on).
3. After tDLLK and tZQinit expire, the device is ready.

5.3 Notes that bite

- The four mode registers are not initialized to defined values by hardware; they must all be written explicitly. Skipping one leaves undefined settings.
- Every MRS command requires all banks precharged and idle, CKE high, tMRD before the next MRS, and tMOD before a non-MRS command such as ZQCL or an ACT.
- The DLL must be enabled (MR1 A0 = 0) before it is reset (MR0 A8 = 1). After any DLL reset, wait tDLLK before issuing a Read or any synchronous ODT operation.
- ZQ calibration initializes the output driver and ODT impedance; it is required before normal operation, not optional.
- During initialization, ODT is not active even if the eventual MR1 setting will enable RTT_NOM. Do not rely on termination until after tDLLK and tZQinit.

Source: JESD79-3F sections 3.3.1, 3.3.2

6 Mode Registers

DDR3 provides four mode registers, selected by the bank-address pins during an MRS command. Their values are undefined after power-up, so all four must be written during initialization. Reprogramming later is allowed whenever all banks are idle.

Register select encoding:

BA2	BA1	BA0	Register
0	0	0	MR0
0	0	1	MR1
0	1	0	MR2
0	1	1	MR3

6.1 MR0 (BA2=BA1=BA0=0)

Field	Bits	Encoding
Burst length	A1-A0	00 = fixed BL8, 01 = on-the-fly BC4/BL8 via A12/BC#, 10 = fixed BC4, 11 = reserved
Read burst type	A3	0 = sequential, 1 = interleaved
CAS latency	A6-A4, A2	see CL table below
Test mode	A7	0 = normal; 1 = manufacturer test only
DLL reset	A8	1 = reset (self-clearing)
Write recovery	A11-A9	see WR table below
Precharge power-down DLL control	A12	0 = slow exit (DLL off), 1 = fast exit (DLL on)

CL encoding:

A6	A5	A4	A2	CL
0	0	1	0	5
0	1	0	0	6
0	1	1	0	7
1	0	0	0	8
1	0	1	0	9
1	1	0	0	10
1	1	1	0	11 (optional at DDR3-1600)
0	0	0	1	12
0	0	1	1	13
0	1	0	1	14

All other A6:A4:A2 combinations are reserved. DDR3 has no half-clock latencies. Overall read latency is RL = AL + CL.

WR encoding:

A11	A10	A9	WR (cycles)
0	0	0	16
0	0	1	5
0	1	0	6
0	1	1	7
1	0	0	8
1	0	1	10
1	1	0	12
1	1	1	14

Program WR to the ceiling of tWR(ns) / tCK(ns). It is used with tRP to determine tDAL for auto-precharge writes.

6.2 MR1 (BA2=0, BA1=0, BA0=1)

Field	Bits	Encoding
DLL enable	A0	0 = enabled (required for normal operation), 1 = disabled
Output driver impedance	A5, A1	00 = RZQ/6, 01 = RZQ/7, others reserved
RTT_NOM (ODT)	A9, A6, A2	000 = off, 001 = RZQ/4, 010 = RZQ/2, 011 = RZQ/6, 100 = RZQ/12, 101 = RZQ/8, others reserved
Additive latency	A4-A3	00 = 0, 01 = CL-1, 10 = CL-2, 11 = reserved
Write leveling	A7	0 = off, 1 = enabled
TDQS enable	A11	0 = off, 1 = enabled (x8 only; disables DM)
Output disable (Qoff)	A12	0 = outputs enabled, 1 = DQ/DQS/DQS# disabled

The DLL is automatically turned off during self-refresh and re-enabled on exit. When the DLL is disabled, synchronous ODT and dynamic ODT are not supported; keep ODT low and set Rtt_WR to off.

6.3 MR2 (BA2=0, BA1=1, BA0=0)

Field	Bits	Encoding
CAS write latency	A5-A3	see CWL table below
Auto self-refresh	A6	0 = manual (use SRT), 1 = ASR enabled (optional)
Self-refresh temperature	A7	0 = normal range, 1 = extended range (optional)
Dynamic ODT (Rtt_WR)	A10-A9	00 = off, 01 = RZQ/4, 10 = RZQ/2, 11 = reserved
Partial array self-refresh	A2-A0	optional; see PASR table below

CWL encoding:

A5	A4	A3	CWL	tCK(avg) range
0	0	0	5	>= 2.5 ns
0	0	1	6	2.5 ns > tCK >= 1.875 ns
0	1	0	7	1.875 ns > tCK >= 1.5 ns
0	1	1	8	1.5 ns > tCK >= 1.25 ns
1	0	0	9	1.25 ns > tCK >= 1.07 ns
1	0	1	10	1.07 ns > tCK >= 0.935 ns
1	1	0	11	0.935 ns > tCK >= 0.833 ns
1	1	1	12	0.833 ns > tCK >= 0.75

A5	A4	A3	CWL	tCK(avg) range
				ns

Overall write latency is WL = AL + CWL.

PASR encoding (optional; data outside selected banks is lost on self-refresh entry):

A2	A1	A0	Array kept active
0	0	0	Full array
0	0	1	Half array (BA 000-011)
0	1	0	Quarter array (BA 000-001)
0	1	1	1/8 array (BA 000)
1	0	0	3/4 array (BA 010-111)
1	0	1	Half array (BA 100-111)
1	1	0	Quarter array (BA 110-111)
1	1	1	1/8 array (BA 111)

6.4 MR3 (BA2=0, BA1=1, BA0=1)

Field	Bits	Encoding
MPR operation	A2	0 = normal DRAM array, 1 = reads redirected to MPR
MPR location	A1-A0	00 = predefined pattern, 01-11 = reserved

When A2 is 0, A1-A0 are ignored. With MPR enabled, only RD/RDA commands are legal until MPR is disabled.

6.5 Programming rules

- Precondition: all banks precharged and idle, tRP satisfied, all bursts finished, CKE high.
- An MRS command rewrites the whole addressed register; always drive every field intentionally, including unchanged ones.
- Wait tMRD between consecutive MRS commands and tMOD before any non-MRS command (NOP/Deselect excluded).
- If RTT_NOM is enabled before or after the MRS, keep ODT low through the command and do not raise it until tMOD expires. If RTT_NOM is disabled, ODT may be either level.
- Drive BA2 and all reserved address bits to 0.
- MRS and DLL reset do not disturb stored array data.

Source: JESD79-3F sections 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5

7 Command Truth Tables

Every command is sampled on a rising CK edge from CS, RAS, CAS, WE, BA, and A. H = high, L = low, X = do not care (may float), V = a valid logic level. CKE transitions add power modes and self-refresh.

7.1 Command truth table

Function	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	A12	BA / address	Notes
MRS	H	H	L	L	L	L	OP	OP	BA = register, opcode on A	
REF	H	H	L	L	L	H	X	X	X	all banks idle
Self-refresh entry	H	L	L	L	L	H	X	X	X	REF encoding + falling CKE
Self-refresh exit	L	H	X	X	X	X	X	X	NOP or DES	asynchronous
Precharge one bank	H	H	L	L	H	L	L	X	BA = bank	
Precharge all	H	H	L	L	H	L	H	X	X	
Activate	H	H	L	L	H	H	row	X	BA = bank, row on A	
Writ	H	H	L	H	L	L	L	V	BA +	A12

Function	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	A12	BA / address	Notes
e fixed BL8 or BC4									column	selects burst length when OTF is enabled
Write BL8 on-the-fly	H	H	L	H	L	L	L	H	BA + column	
Write BC4 on-the-fly	H	H	L	H	L	L	L	L	BA + column	
Write w/ auto-precharge	H	H	L	H	L	L	H	V	BA + column	A10 = 1
Read fixed BL8 or BC4	H	H	L	H	L	H	L	V	BA + column	A12 selects burst length whe

Function	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	A12	BA / address	Notes
										n OTF is enabled
Read BL8 on-the-fly	H	H	L	H	L	H	L	H	BA + column	
Read BC4 on-the-fly	H	H	L	H	L	H	L	L	BA + column	
Read w/ auto-precharge	H	H	L	H	L	H	H	V	BA + column	A10 = 1
NOP	H	H	L	H	H	H	X	X	X	
Deselect	H	H	H	X	X	X	X	X	X	
Power-down entry	H	L	X	X	X	X	X	X	NOP or DES	
Power-down	L	H	X	X	X	X	X	X	NOP or DES	

Function	CKE prev	CKE cur	CS	RAS	CAS	WE	A10	A12	BA / address	Notes
exit										
ZQ calibration long	H	H	L	H	H	L	H	X	X	
ZQ calibration short	H	H	L	H	H	L	L	X	X	
RESET	X	X	X	X	X	X	X	X	RESET# = L	asynchronous, overrides command bus

Key mnemonics:

- ACT = RAS low, CAS and WE high.
- PRE/PREA = RAS low, CAS high, WE low; A10 chooses one bank vs all banks.
- RD = CAS low, WE high; WR = CAS and WE low; A10 adds auto-precharge.
- MRS = all three command strobes low; BA selects MR0-MR3.
- REF = RAS and CAS low, WE high; the same encoding with falling CKE becomes self-refresh entry.
- A12/BC is the burst-length control: 1 = BL8, 0 = BC4. It is ignored when the mode register fixes the burst length.

7.2 CKE truth table

Current state	CKE prev	CKE cur	Command (N)	Result
Power-down	L	L	X	remain in power-down
Power-down	L	H	NOP or DES	exit power-down
Self-refresh	L	L	X	remain in self-refresh
Self-refresh	L	H	NOP or DES	exit self-refresh
Bank(s) active	H	L	NOP or DES	active power-down entry
Reading / writing / precharging	H	L	NOP or DES	power-down entry after burst finishes
All banks idle	H	L	NOP or DES	precharge power-down entry
All banks idle	H	L	REF	self-refresh entry

7.3 Data mask

One DM pin per 8 DQ bits is sampled with the write data. DM = 0 writes the byte; DM = 1 masks it and the byte is not changed. DM is unused during reads. On x8 devices the same ball can be configured as TDQS instead of DM.

7.4 Not in the table

- No Burst Terminate command exists in DDR3 (DDR had one; DDR2 and DDR3 dropped it).
- No Off-Chip Driver Calibration (OCD) command exists in DDR3; OCD was removed.
- No per-bank refresh exists; a REF cycle always refreshes all banks.
- DBI, parity, and ECC are not part of the base DDR3 command set.

Source: JESD79-3F sections 4.1, 4.2, 4.3, 4.4, 4.9

8 Commands: Rules and Constraints

What each command does and what must come next. Timing symbols are defined in Chapter 6; examples use the drill model (B0-B7, R0-R7, C0-C7).

8.1 ACT - Bank Activate

Opens row R in bank B so reads and writes can follow.

- ACT -> RD/WR same bank: wait tRCD, unless additive latency posts the command. With AL > 0 the RD/WR can be issued immediately, but internal CAS must not occur before tRCD.
- ACT -> ACT different bank: tRRD.
- ACT -> ACT same bank: tRC (= tRAS + tRP), and the prior row must already be precharged.
- ACT -> PRE same bank: tRAS (the open row must remain active long enough).
- On 8-bank devices: at most 4 ACTs in any rolling tFAW window.

8.2 RD / RDA - Read

Burst read starting at the given column. Data appears RL = AL + CL clocks after the command. DQS begins toggling one cycle before the first data (read preamble).

- RD -> RD (any bank): tCCD = 4 clocks; back-to-back reads can be issued every tCCD.
- RD -> PRE same bank: earliest spacing is AL + tRTP, also bounded by tRAS. tRTP.MIN = max(4 nCK, 7.5 ns).
- RDA (A10=1): auto-precharge starts internally after the same AL + tRTP delay, delayed if tRAS has not been met. Next ACT to that bank needs tRP from when precharge starts and tRC from the old ACT.
- RD -> WR (book symbol tRTW): the spec does not name tRTW; the figures give the command spacing as RL + BL/2 + 2 - WL clocks. For BL8 this is RL + tCCD + 2 - WL; for BC4 it is RL + tCCD/2 + 2 - WL.

8.3 WR / WRA - Write

Burst write. The first data is captured WL = AL + CWL clocks after the command. DQS must satisfy tDQSS relative to CK.

- WR -> WR (any bank): tCCD = 4 clocks.
- WR -> RD: wait until the last write data has been absorbed and tWTR has passed. tWTR is measured from the first rising CK after the last write data.
- WR -> PRE same bank: WL + BL/2 + tWR. tWR is the write recovery time programmed in MR0; violating it corrupts data.
- WRA (A10=1): internal auto-precharge starts after write recovery. The bank can be re-activated once (WRA + tRP) and tRC are met.
- DM masks individual bytes during writes. DDR3 has no DBI or write parity.

8.4 PRE / PREA - Precharge

Closes the open row and restores data to the array. A10=0 closes the bank selected by BA; A10=1 closes all banks.

- PRE -> ACT same bank: tRP. Unlike DDR2, DDR3 does not add an extra clock for PREA; both use the same tRP.
- Illegal if it would truncate a burst still in flight.

8.5 REF - Refresh

One internal refresh step across all banks; the address counter is internal.

- Precondition: all banks precharged and tRP met.
- REF -> ACT or REF: tRFC (density-dependent).
- Average refresh interval tREFI = 7.8 us at case temperatures up to 85C; 3.9 us in the optional extended range (85C to 95C).
- Scheduling slack: up to 8 REFs may be postponed, so the largest gap between two REFs is 9 x tREFI. Up to 8 REFs may be pulled in early; at most 16 REFs may occur within any 2 x tREFI window.

8.6 Self-refresh entry / exit

Entry: REF encoding with CKE falling, all banks idle, ODT off. The DRAM keeps itself refreshed without needing CK; only CKE matters.

- Minimum stay: tCKE, plus an internal refresh occurs within tCKE of entry.
- Exit: CK must be stable before CKE rises. NOP/DES during the exit window.
- tXS after exit before any non-read command; tXSDLL before any command that needs a locked DLL (such as RD). The DDR3 spec names these tXS/tXSDLL (not tXSR/tXSRD).
- Before re-entering self-refresh, issue at least one REF.
- SRT (MR2 A7) and optional ASR (MR2 A6) select the self-refresh temperature range.

8.7 Power-down

CKE low with NOP/DES. Two flavors: precharge PD (all banks closed) and active PD (a row open).

- Precharge PD has fast exit (MR0 A12 = 1, DLL kept on) and slow exit (MR0 A12 = 0, DLL off).
- Active PD always uses fast exit; the DLL stays on.
- Exit to any valid command: tXP. After slow precharge PD exit, commands needing the DLL (RD, RDA, ODT-controlled transitions) wait tXPDLL.
- CKE low pulse must satisfy tCKE. The CKE transition windows need NOP/DES coverage (tCPDED).
- PD performs no refresh; stay bounded by refresh requirements, at most 9 x tREFI if 8 REFs were posted beforehand.

8.8 NOP and DES

Bus fillers. NOP keeps CS low; DES raises CS. Neither changes state. Use them to occupy forced gaps (tRRD, tCCD, turnaround bubbles, tCPDED windows).

8.9 RESET

RESET# low is an asynchronous override. It resets the device, aborts any operation, and forces outputs to High-Z. After RESET# rises the normal power-up or stable-power reset sequence must be followed (200 us at power-up or 100 us with stable power, then 500 us before CKE high, etc.). ODT is High-Z while RESET# is asserted and until CKE is registered high.

8.10 MRS / mode-register updates

MRS writes MR0-MR3. All address fields of the target register are rewritten, even if only a subset changes.

- MRS -> MRS: tMRD.
- MRS -> any non-NOP/DES command: tMOD.
- MRS must be issued when all banks are idle, all bursts complete, and CKE high. If RTT_NOM is enabled, ODT must be held low before and during the MRS command until tMOD expires.
- DLL reset (MR0 A8 = 1) needs tMOD and then tDLLK before DLL-dependent commands.

8.11 Frequency change in precharge power-down

The input clock frequency may only change during self-refresh or precharge power-down. In precharge PD: hold ODT and CKE low, wait tCKSRE after CKE falls, change frequency, provide a stable clock for tCKSRX, then exit PD. After exit and tXP, reset the DLL via MRS and update CL/CWL/WR as needed. RTT_NOM must be disabled or ODT held low during the frequency change.

8.12 Write leveling entry / exit

Used to align DQS-DQS# to CK-CK# across fly-by routing.

- Entry: MRS to MR1 with A7 = 1, output buffers of other ranks disabled via MR1 A12 = 1. Only NOP/DES and another MRS are valid in leveling mode.
- The DRAM asynchronously feeds back the sampled CK level on DQ. The controller sweeps DQS delay until it sees a 0 -> 1 transition.
- Exit: stop driving DQS, drive ODT low, then MRS MR1 with A7 = 0. Wait tMOD before normal commands; MRS-to-MRS spacing is tMRD.

8.13 MPR read flow

The Multi-Purpose Register outputs a predefined calibration pattern.

- Precharge all banks and wait tRP.
- MRS to MR3 with A2 = 1 (and A[1:0] selecting the pattern) to enter MPR mode.
- Wait tMRD and tMOD.
- Issue RD/RDA with A[1:0] = 00 and A2 = 0 for BL8; only RD/RDA are legal in MPR mode. RDA behaves like a plain read (auto-precharge is ignored).
- After the last MPR burst, wait tMPRR, then MRS to MR3 with A2 = 0 to exit.
- Wait tMRD and tMOD before normal array commands.

Source: JESD79-3F sections 3.3, 3.4.1, 4.7, 4.8, 4.10, 4.11, 4.12, 4.13, 4.14, 4.15, 4.16, 4.17

9 Walkthrough 1: AXI4 Read, Row Hit

9.1 Controller configuration used in all Chapter 4 walkthroughs

Drill model geometry (B0-B7, R0-R7, C0-C7), BL8, additive latency off (AL=0; AL is a programmable field in MR1, but this drill keeps it disabled):

Parameter	Value (clocks)
CL / CWL / AL / RL / WL	5 / 5 / 0 / 5 / 5
tRCD / tRP / tRAS / tRC	5 / 5 / 15 / 20
tRRD / tFAW / tCCD	4 / 16 / 4
tWTR / tWR / tRTP	5 / 10 / 5

The AXI4 data bus is assumed as wide as one DRAM burst, so one AXI4 beat maps to one DRAM burst word: an AXI4 burst of 8 beats is exactly one BL8 DRAM burst. Real controllers slice and dice; the drill model does not. Burst chop (BC4) versus BL8 can be chosen on-the-fly via A12 when MR0 allows it; these walkthroughs keep A12 high for BL8.

9.2 Scenario

Initial bank state: B3 has row R5 open. All other banks idle.

AXI4 master issues: ARADDR mapping to bank B3, row R5, column C2, ARLEN=7 (8 beats), ARSIZE matching full bus width.

9.3 Command trace

```
RD  B3 C2        # row R5 already open in B3 - no ACT needed, go
straight to CAS
```

One command. That is the whole point of a row hit.

Timeline:

```
clk:      0        1    2    3    4    5    6    7    8    9   10   11
12
CMD:      RD
DQS:                        pre  ----DQS toggling----
DQ:                              D0   D1   D2   D3   D4   D5   D6   D7
```

Data appears RL = AL + CL = 5 clocks after the RD command. AL is programmed to zero in this drill, so the CAS is executed immediately; there is nothing to post against. DDR3 computes write latency as AL + CWL, not the RL-1 shortcut used in DDR2.

9.4 What happens on the AXI side

- AR handshake completes; the controller's address mapper splits the address into {B3, R5, C2}.
- The bank machine for B3 reports R5 open -> issue RD immediately, skipping ACT entirely.
- At RL the DRAM drives DQS/DQ; the PHY returns 8 data words; the controller streams them out as 8 AXI R beats with RLAST on the final word.

9.5 Scheduling takeaways

- Row-hit reads are the cheapest DRAM transaction: 1 command, latency RL.
- The bank stays open afterwards (A10=0). A follow-up access to R5 is another hit; an access to a different row of B3 now costs a PRE + ACT (see Walkthrough 2).
- A controller doing FR-FCFS scheduling reorders the queue exactly to manufacture this case.

Source: JESD79-3F sections 3.4.2.1, 3.4.3.4, 3.4.4.2, 4.11, 4.13.1

10 Walkthrough 2: AXI4 Read, Row Miss

Same controller configuration as Walkthrough 1 (CL=5, CWL=5, AL=0, RL=5, WL=5, BL8; tRCD=5, tRP=5, tRAS=15).

10.1 Scenario

Initial bank state: B5 has row R6 open (from earlier traffic). AXI4 read arrives: ARADDR maps to bank B5, row R2, column C1, ARLEN=7.

Row R2 is not the open row -> row miss. The bank must be closed and the right row opened before the read.

10.2 Command trace

```
PRE B5           # close R6 - wrong row is open in B5, precharge
before re-activating
--- (tRP = 5) ---# sense amps need tRP to finish restoring before the
next ACT
ACT B5 R2        # open the requested row R2
--- (tRCD = 5) ---# row decode needs tRCD before a CAS command is
legal
RD  B5 C1        # AL=0: no posted CAS, issue exactly at ACT+tRCD
```

Timeline:

```
clk:      0     1    2    3    4     5     6     7     8     9    10
11   12   13   14   15   16   17   18   19   20
CMD:      PRE                              ACT                    RD
                |<----- tRP=5 ----->|       |<----- tRCD=5 ----->|
DQ:
D0   D1   D2   D3   D4   D5   D6   D7
```

- PRE at clk 0, ACT legal at clk 5 (tRP).
- RD issued at clk 10 (tRCD after ACT at clk 5).
- Read data starts at clk 10 + RL = clk 15.

10.3 Cost accounting

Row miss penalty = tRP + tRCD of pure dead time on this bank before data can flow: 10 clocks here, versus 0 for a hit. With AL=0, posted CAS is off, so the RD command slot is spent at ACT+tRCD; nothing hides the latency.

10.4 AXI side

- The AR channel accepts the request; the controller cannot produce R beats until the DRAM data comes back, so the master just sees a longer RVALID latency. Nothing on the AXI interface changes - all of this is hidden inside the controller.
- If the master had a second read outstanding to a different (idle) bank, a good scheduler would run that one first during the tRP/tRCD bubble.

10.5 Variation: read with auto precharge (RDA)

If the controller predicts no further use of R2 soon, it issues `RDA B5 C1` (A10=1) instead. The bank then precharges itself after the read burst completes and tRTP is satisfied, and is idle for the next tenant without a separate PRE command slot. In DDR3 the minimum read-to-precharge spacing for auto-precharge is AL + tRTP clocks after the RD command, delayed if tRAS is not yet met.

Source: JESD79-3F sections 3.4.2.1, 3.4.3.4, 4.11, 4.12, 4.13.1, 4.13.3

11 Walkthrough 3: AXI4 Write to an Idle Bank

Same controller configuration (CL=5, CWL=5, AL=0, RL=5, WL=5, BL8; tRCD=5, tRP=5, tWR=10).

11.1 Scenario

Initial state: all banks idle. AXI4 write arrives: AWADDR maps to bank B1, row R4, column C0, AWLEN=7, followed by 8 W-channel beats.

An idle bank needs no precharge - miss-to-idle is cheaper than miss-to-open-row by tRP.

11.2 Command trace

```
ACT B1 R4        # open the row; bank was idle so no PRE needed
--- (tRCD = 5) ---# wait for row decode before issuing CAS
WR  B1 C0        # AL=0: issued exactly at ACT+tRCD
--- (WL = 5) --- # write latency: first DQS edge lands 5 clks after
the WR command
--- (tWR = 10) ---# write recovery: last data must soak into the array
before PRE
PRE B1           # close the row (only if the controller wants the
bank back)
```

Timeline:

```
clk:      0     1     2    3    4    5     6    7    8    9   10   11
12   13   14   15   16   17   18   19   20   21   22   23   24
CMD:      ACT                           WR
PRE
                                    |<---- WL=5 ---->|
DQS:                                          pre -DQS toggling-
DQ:                                            D0   D1   D2   D3   D4
D5   D6   D7

|<--- tWR=10 --->|
PRE legal at clk 5 + 5(WL) + 4(BL/2) + 10(tWR) = clk 24 from ACT,
i.e. WL + BL/2 + tWR = 19 clocks after the WR command.
```

11.3 AXI side

- AW and W are independent channels; the controller buffers the 8 W beats (or streams them) while the DRAM side runs ACT + WR.
- BREADY goes back to the master once the controller has accepted responsibility for the data - typically after the write is safely in flight to the DRAM, not after tWR completes.
- Write data is strobed in with DQS driven by the controller/PHY; tDQSS alignment to CK is the PHY's job.

11.4 Write recovery is the trap

tWR is not a turnaround nicety; it is array physics. Precharging B1 before tWR expires destroys the data just written - the bits never finish committing from the sense amps to the cells. This is why WR is a mode-register field: the DRAM itself uses the programmed WR value to time auto precharge after a WRA, and tDAL = WR + roundup(tRP/tCK) is the real “write to next activate” cost when auto precharge is used.

11.5 Comparison against the read miss

	Read miss (walkthrough 2)	Write to idle bank
Commands before data	PRE, ACT, RD	ACT, WR
Dead time before data	tRP + tRCD	tRCD (hidden by WL buffering)
Tail cost	none (bank left open)	tWR before PRE

Source: JESD79-3F sections 3.4.2.1, 3.4.4.2, 4.11, 4.12, 4.14.1, 4.14.2

12 Walkthrough 4: Mixed Traffic and Bus Turnarounds

Same controller configuration (CL=5, CWL=5, AL=0, RL=5, WL=5, BL8; tWTR=5, tWR=10, tCCD=4, tRRD=4, tFAW=16).

The shared DQ bus can only face one way at a time. Read-to-write and write-to-read direction changes cost idle bus clocks - the turnaround bubbles - and they are where bandwidth goes to die in mixed traffic.

12.1 Scenario

Four AXI4 requests, in arrival order:

1. Read B3 R5 C2 (B3 has R5 open - hit)
2. Write B1 R4 C0 (B1 idle)
3. Read B2 R1 C6 (B2 idle)
4. Read B3 R5 C7 (same open row as #1 - hit)

12.2 Command trace

```
RD  B3 C2        # hit: data flows immediately, RL=5
--- (RD->WR: BL/2+2 = 6 clk command gap) ---   # bus must clear the
read burst
ACT B1 R4        # schedule during the turnaround bubble - command bus
is free
WR  B1 C0        # AL=0; first DIN at WR+WL(5); bus turned around
--- (WR->RD: WL+BL/2+tWTR = 14 clk) ---        # last DIN must reach
the array
ACT B2 R1        # again hidden inside the bubble
RD  B2 C6        # next read legal 14 clks after the WR
RD  B3 C7        # hit; tCCD=4 after previous RD - seamless
```

Timeline (clock numbers approximate the spacing rules):

```
clk:    0    1    2    3    4    5    6    7    8    9   10   11   12
13   14   15   16   17   18   19   20   21   22   23
CMD:    RD   ACT                   WR
ACT             RD              RD
             |<- 6 ->|                |<------- 14 ------->|
bus:    ........ D0-D7 ....... p  D0-D7 ............................. p
D0-D7 D0-D7
dir:        RD->      (bubble)    WR->           (bubble)
RD->    RD->
```

12.3 The two turnaround rules

Transition	Command spacing	Why
RD -> WR	BL/2 + 2 (6 clk at BL8)	Read burst + postamble must clear the bus before the write preamble
WR -> RD	WL + BL/2 + tWTR (14 clk here)	tWTR is measured from the first CK edge after the last write data; WL + BL/2 positions that edge from the WR command

The RD->WR spacing is specified by JESD79-3F as RL + tCCD + 2 - WL, which simplifies to BL/2 + 2 when RL = WL. The spec does not name this rule; this book uses tRTW as a common book symbol.

tWTR is internal and invisible on the bus, but the command spacing rule enforces it. CL does not appear in the WR->RD rule: with CWL programmed independently (MR2), the write tail is positioned by WL alone, and RL runs from the RD command itself.

12.4 Scheduling takeaways

- Turnarounds are more expensive than same-direction tCCD gaps: 6 and 14 clocks here versus 4. Grouping reads with reads and writes with writes (write batching / read prioritization) is the single biggest bandwidth lever in mixed traffic.
- ACT commands to other banks are nearly free: the command bus is idle during data turnarounds, so row misses on other banks hide completely.
- tFAW still applies: this trace's two ACTs (clk 1 and clk 19) are 18 clocks apart, fine; a burst of five ACTs within 16 clocks would stall the fifth.
- With only 8 banks in the drill model, tFAW (4 ACTs per 16 clocks) can bind before tRRD does on an activate-heavy stream - watch both.
- Maintenance commands such as ZQCS/ZQCL, REF, or any DLL re-lock step are all-bank and require the channel to be idle; a controller must reserve slots for them, adding further bubbles in sustained traffic.

12.5 AXI side

The AXI fabric never sees any of this. Read masters see variable RVALID latency; write masters see variable WREADY/BVALID timing. A controller with enough queue depth reorders requests 1-4 (respecting AXI ordering rules per ID) to convert the trace above into two reads back-to-back and the write parked until the bus is free.

Source: JESD79-3F sections 3.4.2.1, 3.4.3.4, 3.4.4.2, 4.11, 4.12, 4.13.1, 4.14.1, 4.14.2, 5.5

13 Data Path: Prefetch, Bursts, Strobes

13.1 8n prefetch

Each column access fetches eight words per IO group between the array and the IO buffers in one internal transfer. The array core therefore runs at one eighth of the pin data rate. This is the architectural jump from DDR2 to DDR3: the external bus doubled again while the core clock stayed behind.

Consequences that shape the rest of the protocol:

- BL8 is the natural burst; BC4 (burst chop) is a BL8 transfer truncated to four words on the bus.
- Timing measured from internal data movement is anchored to the last 4-word half of the burst, which is why BC4 and BL8 sometimes differ by 2 clocks in rules such as tWR and tWTR.
- The array only supports aligned 8-word transfers; the controller must still respect burst boundaries.

13.2 Burst length and burst order

DDR3 supports BL8 and BC4, selected either by MR0 bits A1:A0 or on-the-fly via A12/BC# during each READ or WRITE command. In the on-the-fly mode, A12=1 selects BL8 and A12=0 selects BC4. A12 is used only for burst control, not as a column address.

MR0 A1:A0	A12 (OTF)	Burst
00	1	BL8 (fixed)
01	0/1	BL8 or BC4 selected via A12
10	0	BC4 (fixed)
11	-	reserved

Burst type (MR0 A3) selects sequential or interleaved address order. The table below shows the data word order inside the burst.

BL	Start A2:A0	Sequential	Interleaved
BC4 (read)	000	0,1,2,3	0,1,2,3
BC4 (read)	001	1,2,3,0	1,0,3,2
BC4 (read)	010	2,3,0,1	2,3,0,1
BC4 (read)	011	3,0,1,2	3,2,1,0
BC4 (read)	100	4,5,6,7	4,5,6,7
BC4 (read)	101	5,6,7,4	5,4,7,6
BC4 (read)	110	6,7,4,5	6,7,4,5
BC4 (read)	111	7,4,5,6	7,6,5,4
BC4 (write)	0,V,V	0,1,2,3	0,1,2,3
BC4 (write)	1,V,V	4,5,6,7	4,5,6,7
BL8 (read/write)	000	0,1,2,3,4,5,6,7	0,1,2,3,4,5,6,7
BL8 (read/write)	001	1,2,3,0,5,6,7,4	1,0,3,2,5,4,7,6
BL8 (read/write)	011	3,0,1,2,7,4,5,6	3,2,1,0,7,6,5,4
BL8 (read/write)	101	5,6,7,4,1,2,3,0	5,4,7,6,1,0,3,2
BL8	111	7,4,5,6,3,0,1,2	7,6,5,4,3,2,1,0

BL	Start A2:A0	Sequential	Interleaved
(read/write)			

For writes, only A2 is meaningful because the lower column bits are ignored. When burst length is fixed to BC4 by MR0, the internal write starts two clocks sooner than in BL8 mode, pulling in tWR and tWTR. When BC4 is selected OTF, the internal timing follows BL8.

13.3 DQS: the source-synchronous strobe

DQ data is referenced to DQS, not directly to CK.

- Reads: the DRAM drives DQS edge-aligned with DQ. The read preamble (tRPRE, minimum 0.9 tCK) places DQS low before the first data; the postamble (tRPST, minimum 0.3 tCK) marks the end. tDQSCK bounds the DQS rising edge against CK, tDQSQ bounds the latest valid DQ transition against DQS, and tQH bounds the earliest invalid transition. tQSH and tQSL describe the high and low pulse widths of the differential strobe.
- Writes: the controller drives DQS and must keep it within tDQSS of CK (roughly +/-0.25 tCK). DQ setup and hold against DQS are tDS and tDH. The write preamble tWPRE and postamble tWPST frame the burst; both have minimum values of 0.9 tCK and 0.3 tCK respectively.
- DQS is normally differential (DQS/DQS#). DDR3 also defines TDQS/TDQS# on x8 parts as a termination reference for the strobe when enabled in MR1.

13.4 Write leveling

Fly-by module routing creates different CK-to-DQS delays at each DRAM on a DIMM. To let the controller compensate, DDR3 provides write leveling. The controller sets MR1 A7=1 to enter leveling mode, then sweeps DQS delay while the DRAM feeds back the sampled CK level on the DQ pins. The controller stops when a 0->1 transition is seen, establishing a DQS-to-CK relationship that satisfies tDQSS, tDSS and tDSH. Only the DQS/DQS# terminations are controlled by the ODT pin in this mode; normal read/write decode is suspended. Exit is via another MRS that clears MR1 A7.

13.5 Write data mask (DM)

One DM pin per byte lane is sampled with write data; a high level masks that byte so the addressed cell keeps its old value. DM is not used during reads. On x8 devices, the same pin may be repurposed as TDQS/TDQS# by MR1 A11, in which case byte-level write masking is unavailable. There is no data bus inversion in DDR3.

13.6 Latency arithmetic, collected

Quantity	Formula	Programmed by
RL (read latency)	AL + CL	MR1 (AL), MR0 (CL)
WL (write latency)	AL + CWL	MR1 (AL), MR2 (CWL)
RD cmd -> data	RL clocks	
WR cmd -> first DQS	WL clocks	
RD -> RD, WR -> WR	max(tCCD=2, BL/2)	
RD -> WR	BL/2 + 2 clocks (book symbol tRTW)	

Source: JESD79-3F sections 3.4.2, 3.4.3, 3.4.4, 4.8, 4.13, 4.14

14 ODT - On-Die Termination

14.1 What it is

DDR3 keeps termination on the die, but adds a second programmable value. MR1 holds the normal termination strength Rtt_Nom; MR2 holds a separate write termination strength Rtt_WR. During writes the device can switch from Rtt_Nom to Rtt_WR automatically without a new MRS command. This is dynamic ODT, and it is the big difference from DDR2.

The ODT pin still enables and disables termination cycle by cycle. When any of MR1 {A9,A6,A2} or MR2 {A10,A9} are non-zero, ODT mode is enabled; otherwise the DRAM ignores the ODT pin.

14.2 Programming

MR1 bits A9:A6:A2 select Rtt_Nom. RZQ is the external 240 ohm reference resistor connected to the ZQ pin.

A9 A6 A2	Rtt_Nom
0 0 0	disabled
0 0 1	RZQ/4 = 60 ohm
0 1 0	RZQ/2 = 120 ohm
0 1 1	RZQ/6 = 40 ohm
1 0 0	RZQ/12 = 20 ohm
1 0 1	RZQ/8 = 30 ohm
1 1 0	reserved
1 1 1	reserved

If Rtt_Nom is used during writes, only RZQ/2, RZQ/4 and RZQ/6 are allowed.

MR2 bits A10:A9 select Rtt_WR:

A10 A9	Rtt_WR
0 0	dynamic ODT disabled
0 1	RZQ/4 = 60 ohm
1 0	RZQ/2 = 120 ohm
1 1	reserved

Dynamic ODT requires the DLL to be on and locked. It is not available during write leveling or DLL-off mode.

14.3 ODT timing

In synchronous ODT mode (DLL on and locked), the ODT pin is sampled on rising CK edges. Termination turns on ODTLon clocks after ODT is sampled high and turns off ODTLoff clocks after ODT is sampled low, where ODTLon = ODTLoff = WL - 2. With additive latency, this becomes CWL + AL - 2.

Symbol	Meaning
ODTLon	ODT high sample -> termination begins to turn on
ODTLoff	ODT low sample -> termination begins to turn off
tAONmin/max	Rtt leaves high-Z / reaches full Rtt, measured from ODTLon
tAOFmin/max	Rtt starts to turn off / reaches high-Z, measured from ODTLoff
ODTH4	minimum ODT high time after assertion or after a BC4 write = 4 tCK
ODTH8	minimum ODT high time after a BL8 write = 6 tCK

Dynamic ODT adds three more latencies measured from the write command:

Symbol	Meaning
ODTLcnw	Rtt_Nom -> Rtt_WR switch = WL - 2
ODTLcwn4	Rtt_WR -> Rtt_Nom for BC4 = 4 + ODTLoff
ODTLcwn8	Rtt_WR -> Rtt_Nom for BL8 = 6 + ODTLoff
tADC	skew window for the Rtt change = 0.3 to 0.7 tCK

14.4 System rules

- Reads: the DRAM cannot terminate and drive at the same time, so ODT must be driven low at least half a clock before the read preamble and may only rise again after the postamble.
- Power-down: synchronous ODT applies in active and idle power-down; in precharge power-down the behavior depends on whether the DLL is kept enabled via MR0 A12. In DLL-off mode ODT must be disabled by holding ODT low and/or clearing Rtt_Nom.
- Reset: the reset pin forces the device into a known state; ODT is off during reset and re-enabled later under normal MRS programming.
- Rtt changes by MRS require a quiet window. After any MRS that affects termination, wait tMOD before the new value is reliable.

14.5 ZQ calibration

The ZQCL command performs a long calibration of output driver and ODT values. It is used at initialization and is allowed tZQinit for the first command after reset and tZQoper for later long calibrations. The ZQCS command performs a short periodic calibration and is allowed tZQCS. During any ZQ calibration the channel must be quiet, CKE must be high, ODT disabled, all banks precharged with tRP met, and all DQ pins high-impedance. ZQ calibration may be issued in parallel with DLL lock time when exiting self-refresh, but an explicit command is required after self-refresh exit (earliest time tXS).

14.6 What DDR3 does NOT have

Stated plainly, because later generations add these and it is easy to back-port them mentally:

- No DBI (data bus inversion). Every DQ drives its true value; there is no inversion flag pin.
- No write or command parity. A corrupted command or write is silently executed.
- No on-die ECC and no ECC storage mode. System-level ECC, if wanted, is the controller's job with extra DRAM devices for check bits.
- No CRC on the data bus (that arrives with DDR4).
- No OCD (off-chip driver calibration). ZQ calibration replaces that idea.

If your design needs any of these properties, build them above the DRAM.

Source: JESD79-3F sections 3.4.3, 3.4.4, 3.4.5, 4.8, 5.1, 5.2, 5.3, 5.5

15 Core Timing Parameters

The bank-state timings: what each parameter constrains, and the command transition it applies to. Row-cycle values live in the speed-bin tables (Tables 62-67, one per DDR3 grade); the ns figures below cite the range across bins. Clock conversion: divide by tCK(avg) and round up (RU).

15.1 Row-cycle parameters

Symbol	Definition	Applies between	Min (typ bins)
tRCD	Row-to-column delay: the row must reach the sense amps before a column command executes	ACT(bank x) -> RD/WR(bank x)	10-15 ns by bin (12.5 at DDR3-800D, 10 at DDR3-1600G)
tRP	Precharge recovery: sense amps and bitlines restored before re-activation	PRE(bank x) -> ACT(bank x)	10-15 ns, bin-matched to tRCD
tRAS	Minimum row-open time: data must be restored to the cells before precharge	ACT(bank x) -> PRE(bank x)	36-37.5 ns; max 9 x tREFI
tRC	Row cycle: full open-close-reopen loop for one bank	ACT(bank x) -> ACT(bank x)	46.5-52.5 ns by bin

Unlike DDR2 there is no extra clock for precharge-all: PREA and PRE both use the same tRP in DDR3.

15.2 Activate-rate parameters (cross-bank)

Symbol	Definition	Applies between	Min
tRRD	Activate-to-activate spacing, any two banks (limits peak array current)	ACT(bank x) -> ACT(bank y)	max(4 nCK, X ns): 10 ns at DDR3-800, 7.5 at 1066, 6 at 1333/1600, 5 at 1866/2133 (1 KB pages; 2 KB pages 2.5 ns looser below 1333)
tFAW	Four-activate rolling window: at most 4 ACTs inside any window of this length	any 4 ACTs within a sliding window	1 KB page: 40/37.5/30/30/27/25 ns at 800/1066/1333/1600/1866/2133; 2 KB page: 50/50/45/40/35/35 ns

tFAW example with the drill model: if RU(tFAW/tCK) = 16 and an ACT lands at clock N, at most three more ACTs may land in clocks N+1 .. N+15.

15.3 Data-to-precharge parameters

Symbol	Definition	Applies between	Min
tRTP	Read-to-precharge: internal analog read must finish before the row closes	internal RD(bank x) -> PRE(bank x)	max(4 nCK, 7.5 ns)
tWR	Write recovery: last write data must be committed to the cells before precharge	first CK after last write data(bank x) -> PRE(bank x)	15 ns; programmed in clocks via MR0 (WR codes 5, 6, 7, 8, 10, 12, 14, 16)
tDAL	Write auto-precharge total: programmed WR plus precharge, before re-activation	WRA(bank x) -> ACT(bank x)	WR + RU(tRP/tCK(avg))

tRTP and tWR are referenced from the first rising CK edge after the last data of the burst - not from the command. tWR grows slower than tCK shrinks, so it costs more clocks at every new speed grade.

Command-spacing formulas that fall out of these (same bank):

Transition	Earliest spacing (clocks)
RD -> PRE	AL + max(RU(tRTP), 4) - with tRTP's 4-nCK floor this is AL + RU(tRTP)
WR -> PRE	WL + BL/2 + RU(tWR)
RDA -> ACT	RU(tRP) after the auto-precharge actually starts (AL + BL/2 + RU(tRTP) after the RDA, delayed if tRAS is unmet), and >= tRC from the old ACT
WRA -> ACT	WL + BL/2 + WR + RU(tRP), and >= tRC

15.4 Why these four numbers dominate

tRCD and tRP are the row-miss penalty. tRAS + tRP = tRC bounds how fast one bank can cycle rows. tRRD and tFAW bound how fast the whole device can open rows regardless of how many banks are free - on activate-heavy streams (many small transfers, open-page policy) tFAW is usually the binding constraint, not tRRD. The controller's scheduler exists to hide tRCD/tRP behind other banks' data and to pace ACTs against tRRD/tFAW.

Source: JESD79-3F sections 4.11-4.14, 12.3 (Tables 62-67), 13.1 (Table 68)

16 Turnaround and Burst Timing

Column-command spacing and bus-direction rules. These are the parameters that decide how full the DQ bus can be kept.

16.1 CAS-to-CAS and latencies

Symbol	Definition	Applies between	Value
tCCD	Column command to column command, same or different banks	RD/WR -> RD/WR (any direction, same type)	4 clocks
RL	Read latency: command to first data	RD -> first DQS/DQ out	AL + CL clocks
WL	Write latency: command to first strobe	WR -> first DQS in	AL + CWL clocks
CL	CAS latency component of RL	internal CAS -> data	5-11 clocks (MR0; CL 5/6 share an encoding bit pattern with A2)
CWL	CAS write latency component of WL	internal write -> first DIN	5-8 clocks (MR2)
AL	Additive latency: posted-CAS hold	RD/WR command -> internal CAS	0, CL-1, or CL-2 (MR1)

BL/2 clocks of bus occupancy per burst: BC4 occupies 2 clocks of DQ, BL8 occupies 4. Seamless same-direction traffic means issuing a column command every BL/2 clocks - at BL8 that is exactly tCCD = 4, so back-to-back BL8 bursts just saturate the bus.

Note the DDR3 change from DDR2: WL is programmed independently via CWL, not derived as RL - 1. A controller may run CL = 9 with CWL = 7; the write tail no longer slavishly follows the read latency.

16.2 Direction changes

Note on tRTW: JESD79-3F never names the read-to-write parameter (the symbol does not appear in the spec). It does specify the spacing: RL + tCCD + 2 - WL for BL8, and RL + tCCD/2 + 2 - WL when the read is chopped to BC4. This book calls the parameter tRTW, matching the other books in this series, which keeps the timing tables comparable across technologies.

Symbol	Definition	Applies between	Value
tRTW	Read-to-write turnaround: read burst and postamble must clear before the write preamble (book symbol; the spec gives the relation unnamed)	RD(any bank) -> WR(any bank)	RL + tCCD + 2 - WL at BL8: 6 clocks when RL = WL
tWTR	Write-to-read: last write data must leave the input path before a read reuses the array	first CK after last write data(any) -> internal RD(any)	max(4 nCK, 7.5 ns)

Command-spacing forms:

Transition	Spacing from first command (clocks)
RD -> WR	RL + tCCD + 2 - WL (= BL/2 + 2 when RL = WL)
WR -> RD	WL + BL/2 + RU(tWTR)
RD -> RD	max(tCCD, BL/2)
WR -> WR	max(tCCD, BL/2)

Worked example (the Chapter 4 configuration: CL = 5, CWL = 5, AL = 0, so RL = WL = 5; BL8; tWTR = 5 clk): RD->WR = 4 + 2 = 6 clocks, WR->RD = 5 + 4 + 5 = 14 clocks. The asymmetry is the write path's long internal tail (WL plus the tWTR analog window); reads turn into writes far cheaper than writes turn into reads - the opposite ranking from the latency numbers alone would suggest.

16.3 Strobe-level timing (for completeness)

These bound the DQS/DQ relationship rather than the scheduler, but the controller's PHY must meet them:

Symbol	Definition	Value
tDQSS	Write DQS rising edge to CK alignment	+/- 0.25 tCK (+/- 0.27 at DDR3-1600 and above)
tWPRE / tWPST	Write DQS preamble / postamble	0.9 min / 0.3 min tCK
tRPRE / tRPST	Read DQS preamble / postamble	0.9 min / 0.3 min tCK
tDQSCK	Read DQS output access from CK (DLL window)	400 ps max at 800, down to 180 ps at 2133
tDQSQ	DQ skew against DQS, per group	200 ps max at 800, down to 75 ps at 1866/2133
tQH	DQ/DQS output hold from DQS	0.38 tCK
tDS / tDH	Write DQ setup / hold against DQS, base values	AC150 / DC100 rows by speed (AC135 for 1866/2133); both derated by slew rate per 13.6.1

16.4 Burst chop and interrupt rules

- BC4 (burst chop): a BL8 fetch truncated to 4 data words, selected by A12 at the command (on-the-fly) or fixed in MR0. tCCD between a BC4 and a following BC4 is 4 clocks; a chopped burst still pays the full fetch.
- BC4 bursts are never interruptible. BL8 bursts may only be cut by a same-direction burst aligned to the 4-word (BC4) boundary.
- There is no Burst Terminate command. A burst runs its full length unless legally interrupted as above.

Source: JESD79-3F sections 4.13, 4.14, 13.1 (Tables 68-69), 13.6.1

17 Refresh and Self-Refresh Timing

17.1 Refresh parameters

DRAM cells leak; every row must be refreshed within the retention window. Refresh is all-banks and internally addressed - the controller only schedules it. DDR3 has no per-bank refresh command.

Symbol	Definition	Applies between	Value
tREFI	Average interval between refresh commands	REF -> REF (average)	7.8 us at case 0-85 C; 3.9 us at 85-95 C (extended-temperature parts)
tRFC	Refresh cycle time: refresh in progress, no other command	REF -> ACT, REF -> REF	90 ns (512 Mb), 110 (1 Gb), 160 (2 Gb), 260 (4 Gb), 350 (8 Gb)

Rules:

- All banks must be precharged (and tRP met) before REF.
- Up to 8 refresh commands may be postponed, so the maximum interval between two surrounding REFs is 9 x tREFI.
- Up to 8 may be pulled in; at most 16 REFs may fall in any 2 x tREFI window.
- Violating refresh timing silently corrupts data; the spec requires that data be rewritten before any valid read.
- tRFC grows steeply with density because one REF must touch more array. At 8 Gb, refresh duty cycle = tRFC/tREFI is about 4.5% of all bus time.

17.2 Self-refresh

The DRAM refreshes itself with the external clock ignored: for sleep states where the controller is powered down but memory contents must survive. DDR3 renames the exit parameters (tXS/tXSDLL where DDR2 said tXSNR/tXSRD).

Symbol	Definition	Applies between	Value
tCKE	Minimum CKE pulse width, high or low	any CKE pulse; SRE -> SRX floor	max(3 nCK, 7.5 ns at 800, 5 ns at 1600+)
tCKESR	Minimum CKE low width for self-refresh entry to exit	SRE -> SRX	tCKE(min) + 1 nCK
tXS	Exit self-refresh to commands not requiring a locked DLL	SRX (CKE high) -> ACT, PRE, MRS, REF	max(5 nCK, tRFC(min) + 10 ns)
tXSDLL	Exit self-refresh to commands requiring a locked DLL	SRX -> RD	tDLLK(min) = 512 nCK
tCKSRE	Valid clock required after self-refresh entry (and PD entry)	SRE -> clock may stop	max(5 nCK, 10 ns) of stable clock
tCKSRX	Valid clock required before self-refresh exit (and PD exit, reset exit)	stable clock -> SRX/PDX	max(5 nCK, 10 ns)

Entry checklist: all banks idle, ODT off (low), REF encoding with CKE falling. The DLL shuts down automatically.

Exit checklist: stable clock first (tCKSRX), then CKE high; NOP/DES through the exit window; wait tXS for non-read commands or tXSDLL for reads (the DLL is re-locking - same 512-clock rule as init); keep ODT off until the exit window completes.

The refresh rate inside self-refresh is set by MR2: ASR (A6) auto-selects from an internal temperature sensor where supported; SRT (A7) forces the doubled 85-95 C rate. Above 85 C with SRT/ASR off, retention is not guaranteed. If the part supports the extended temperature range at all, normal-operation tREFI halves to 3.9 us.

17.3 Power-down timings (related CKE rules)

Symbol	Definition	Value
tXP	Exit power-down (DLL on) to any valid command; also precharge-PD-with-frozen-DLL to non-DLL commands	max(3 nCK, 7.5 ns at 800/1066, 6 ns at 1333+)
tXPDLL	Exit precharge power-down with DLL frozen to DLL-requiring commands	max(10 nCK, 24 ns)
tCPDED	Command pass disable delay after PD entry	1 nCK (2 at 1866/2133)
tPD	Power-down entry to exit timing	tCKE(min) to 9 x tREFI
tACTPDEN / tPRPDEN	ACT / PRE or PREA to PD entry	1 nCK (2 at 1866/2133)
tRDPDEN	RD/RDA to PD entry	RL + 4 + 1 nCK
tWRPDEN	WR to PD entry (BL8/OTF/BC4OTF)	WL + 4 + RU(tWR/tCK)
tWRAPDEN	WRA to PD entry (BL8/OTF/BC4OTF)	WL + 4 + WR + 1
tMRSPDEN	MRS to PD entry	tMOD(min)

Fast vs slow exit is the MR0 A12 choice for precharge power-down: fast exit keeps the DLL running for a tXP-class resume; slow exit freezes the DLL and pays tXPDLL on the way out. Active power-down always uses fast exit. Power-down performs no refresh: duration is bounded by the refresh requirements, 9 x tREFI at best with 8 REFs posted beforehand. In power-down with the DLL frozen, ODT turns asynchronous (tAONPD/tAOFPD, 2-8.5 ns).

Source: JESD79-3F sections 4.15-4.17, 12.2 (Table 61), 13.1 (Table 68)

18 Package and Electrical Notes

A light pass: what matters when connecting a DDR3 device, deliberately skipping the full ballout grids (consult the spec figures for those).

18.1 Packages

- Monolithic parts use FBGA per JEDEC MO-207: dedicated ballouts for x4 (sec 2.1), x8 (sec 2.2) and x16 (sec 2.3) organizations. Unused grid positions are no-connect mechanical support balls; whether they are populated depends on the vendor's package size, so land patterns should assume the maximum allowed package.
- Dual-die and quad-die stacked packages present 2 or 4 ranks behind one channel: each rank gets its own chip select and ODT association, and shares the address/command and data busses.
- New pins versus DDR2: RESET# (full-chip reset, active low) and the ZQ calibration reference pin (bonds to an external 240 ohm resistor).

18.2 Supplies and signaling

Rail	Nominal	Feeds
VDD	1.5 V +/- 0.075	core and DLL (DDR3 drops DDR2's separate VDDL)
VDDQ	1.5 V +/- 0.075	output drivers
VREFCA	0.5 x VDD	command/address input reference
VREFDQ	0.5 x VDDQ	data input reference
VTT	0.75 V nominal	board-side termination rail for the CA bus

- SSTL_15 signaling: single-ended inputs compare against the VREF rails; CK/CK#, DQS/DQS# run differential.
- The CA bus is source-terminated point-to-point or short-stub with VTT pull-ups; the DQ group leans on ODT instead (Chapter 5).
- Case temperature: 0-85 C standard operation; parts rated for the extended range run 85-95 C at half tREFI with the SRT/ASR mode bits set (MR2).

18.3 Speed bins

Grade	CL-nRCD-nRP bins	tRCD/tRP (ns)
DDR3-800	D: 5-5-5, E: 6-6-6	12.5 / 15
DDR3-1066	E: 6-6-6, F: 7-7-7, G: 8-8-8	11.25 / 13.125 / 15
DDR3-1333	F: 7-7-7, G: 8-8-8, H: 9-9-9, J: 10-10-10 (opt)	10.5 / 12 / 13.5 / 15
DDR3-1600	G: 8-8-8, H: 9-9-9, J: 10-10-10, K: 11-11-11 (opt)	10 / 11.25 / 12.5 / 13.75
DDR3-1866 / 2133	per Tables 66-67	13.125+ / see sec 13.2

Every grade carries its own tCK(avg) window per supported CL setting (Tables 62-67): roughly 3.3 ns at the slow end (DLL-off floor 8 ns) down to 1.25 ns at DDR3-1600 and above. All bins share the same core analog floors (tRCD/tRP/tRAS in ns), which is why higher-speed parts need more clocks for the same nanoseconds - and why a controller should store timings in ns and convert to clocks at configuration time.

18.4 Characterization

Section 10 defines the IDD/IDDQ measurement conditions (the familiar current-state table per operation); section 11 gives pin capacitance budgets. Neither changes how the device is scheduled; consult them when sizing the power delivery rather than the scheduler.

Source: JESD79-3F sections 2, 6, 12.3 (Tables 62-67)

19 Test and Calibration Features

DDR3 adds several calibration and diagnostic aids that DDR2 lacked: write leveling, the Multi-Purpose Register (MPR), ZQ calibration, and self-refresh temperature selection.

19.1 Write leveling

Because command/address/clock routing on DDR3 modules uses fly-by topology, CK arrives at each DRAM at a different time relative to its DQS. Write leveling lets the controller align DQS to CK per rank.

Enablement and rules:

- Enter by loading MR1 with A7 = 1. All banks must be idle first.
- Exit by loading MR1 with A7 = 0. After the exit MRS, observe tMOD (and tMRD for further MRS commands) before normal traffic.
- Only NOP, DESELECT, and MRS commands are legal in leveling mode; the allowed MRS changes are the Qoff bit (A12) and the exit bit (A7).
- Other ranks are typically quieted by setting their MR1 A12 = 1, disabling their output buffers so only the target rank feeds back.
- While leveling is active, asserting ODT turns on termination for DQS/DQS# only; DQ termination remains off.

Procedure outline:

```
MRS MR1 A7=1                    # enter write leveling
wait tMOD, then assert ODT
wait tWLDQSEN, then drive DQS low / DQS# high
after both tDQSL and tWLMRD are satisfied, send one DQS/DQS# edge
DRAM samples CK with the rising DQS edge and returns the CK value on
DQ
repeat while adjusting DQS delay until a 0 -> 1 transition is found
stop driving DQS, deassert ODT, MRS MR1 A7=0  # exit
wait tMOD before the next valid command
```

Key timing symbols:

Symbol	Meaning
tWLDQSEN	Wait from ODT assertion until the DRAM has turned on DQS termination
tWLMRD	Wait from the entry MRS until the first DQS/DQS# leveling edge
tDQSL / tDQSH	Minimum low/high pulse width for the leveling strobe edge
tWLO	Delay from the leveling DQS edge to valid DQ feedback
tWLOE	Allowed spread between the earliest

Symbol	Meaning
	and latest DQ transitions during feedback
tMOD	Mode-register command update delay after the exit MRS

The controller adjusts the DQS delay so the DQS rising edge lines up with the CK rising edge at each DRAM, satisfying tDQSS, tDSS, and tDSH.

19.2 Multi-Purpose Register (MPR)

The MPR is a read-only register that sources a fixed calibration pattern. It is used for read-timing training and debug, not for normal data access.

Enablement and rules:

- Load MR3 with A2 = 1 to enter MPR mode. Before that MRS, all banks must be idle and tRP must be satisfied.
- MR3 A[1:0] selects the location; only `00b` (predefined calibration pattern) is valid; the other encodings are RFU.
- In MPR mode the command set shrinks to RD and RDA only. RDA is treated as a plain read; its auto-precharge request is ignored. Writes, power-down entry, and self-refresh entry are not allowed.
- Issue MPR reads only after the DLL has locked.

Predefined pattern readout:

Burst setting	A2	Burst order	Data pattern
BL8	0	0,1,2,3,4,5,6,7	0,1,0,1,0,1,0,1
BC4 lower nibble	0	0,1,2,3	0,1,0,1
BC4 upper nibble	1	4,5,6,7	0,1,0,1

Address rules during MPR reads: drive A[1:0] = `00b`; A2 chooses the nibble for BC4; A12/BC selects burst-chop on-the-fly; all other address and bank bits are don't-care. DQ[0] carries the pattern, and the remaining DQ pins either repeat DQ[0] or drive 0.

Relevant timings: tRP before entry, tMRD/tMOD around the enabling and disabling MRS commands, normal read latency (RL = AL + CL), and tMPRR = 1 nCK after the last MPR burst before exiting MPR mode.

19.3 ZQ calibration

ZQ calibration trims the output driver impedance (Ron) and on-die termination (ODT) against an external reference.

Symbol	Definition	Typical value
tZQinit	First ZQCL after reset/initialization	max(512 nCK, 640 ns)
tZQoper	Subsequent ZQCL during normal operation	max(256 nCK, 320 ns)
tZQCS	Short ZQCS for periodic calibration	max(64 nCK, 80 ns)

- A precision 240 ohm resistor (RZQ) connects the ZQ ball to ground; it may be shared between two DRAMs only if their calibration windows do not overlap.
- ZQCL performs the full calibration; ZQCS performs a shorter correction for voltage and temperature drift.
- Both require all banks precharged and tRP satisfied. The channel must be quiet during the calibration window.
- A ZQCL is required during initialization. Periodic ZQCS commands are recommended afterwards; the interval depends on system temperature and voltage drift rates.
- After self-refresh exit, an explicit ZQ command is required if calibration is needed; the earliest issue time is tXS.
- If multiple DRAMs share one ZQ resistor, their calibration windows must not overlap.

19.4 Self-refresh temperature (SRT / ASR)

Self-refresh timing is configured in MR2:

MR2 A6	MR2 A7	Meaning
0	0	Manual SRT, normal temperature range
0	1	Manual SRT, extended temperature range (only if the part supports it)
1	0	Auto Self-Refresh (ASR) enabled; DRAM manages its own refresh rate
1	1	Illegal

- A6 selects ASR when the device supports it; A7 selects extended-range manual mode when ASR is off.
- If the part does not support extended temperature range, A7 must stay 0.
- Refer to the data sheet or SPD to discover which options are actually implemented.

19.5 Test mode

MR0 A7 selects a vendor-specific test mode. The only user requirement is to keep it 0. No standard user-visible test mode exists at that bit.

19.6 What is not here

- No OCD (off-chip driver calibration) command set: DDR3 replaces it with ZQ calibration.
- No boundary-scan / JTAG test port inside the DRAM device.
- No writes to the MPR; the pattern is fixed and read-only.
- No per-pin loopback test mode.
- No on-die repair readback or post-package repair interface exposed to the user.

Source: JESD79-3F sections 3.4.2.3, 4.8, 4.9, 4.10, 5.5